

CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 12: Surface Areas and Volumes

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Two cubes, each of volume 64 cm^3 , are joined end to end. Find the surface area of the resulting cuboid.
2. A vessel is a hollow hemisphere mounted on a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find its inner surface area.
3. A toy is a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find its total surface area.
4. A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid.
5. A solid is a cone standing on a hemisphere, both of radius 1 cm, and the height of the cone equals its radius. Find the volume of the solid in terms of π .
6. A cylinder of height 10 cm and radius 6 cm has a conical cavity of height 8 cm and the same radius hollowed out. Find the volume of the remaining solid in terms of π .
7. A tent is a cylinder of height 2.1 m and diameter 4 m with a conical top of slant height 2.8 m. Find the area of canvas used.
8. A solid iron pole is a cylinder of height 220 cm and base diameter 24 cm, surmounted by another cylinder of height 60 cm and radius 8 cm. Find the volume of the pole. Take $\pi = 3.14$.
9. A sphere of radius 3 cm is dropped into a cylindrical vessel of radius 6 cm that is partly filled with water. If the sphere is fully immersed, by how much does the water level rise?
10. A wooden article is made by scooping out a hemisphere from each end of a solid cylinder of height 10 cm and base radius 3.5 cm. Find the total surface area of the article.

Answer Key

1. Side 4 cm; cuboid $8 \times 4 \times 4$; surface area
= 160 cm^2
2. $2\pi r(h + r) = 2 \times \frac{22}{7} \times 7 \times 13 = 572 \text{ cm}^2$
3. $l = 12.5 \text{ cm}$; $137.5 + 77 = 214.5 \text{ cm}^2$
4. 7 cm; $6 \times 49 + \frac{22}{7} \times 3.5^2 = 332.5 \text{ cm}^2$
5. $\frac{1}{3}\pi + \frac{2}{3}\pi = \pi \text{ cm}^3$
6. $360\pi - 96\pi = 264\pi \text{ cm}^3$
7. $\pi r(2h + l) = \frac{22}{7} \times 2 \times 7 = 44 \text{ m}^2$
8. $3.14 \times (31680 + 3840) = 111532.8 \text{ cm}^3$
9. $\frac{36\pi}{36\pi} = 1 \text{ cm}$
10. $2\pi rh + 4\pi r^2 = 220 + 154 = 374 \text{ cm}^2$